MARK SCHEME · TEACHER COPY

10BR · Chapter 6: Automated and Emerging Technologies

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 11, original Q9 (a),(b),(c) [5 marks]

Question	Answer	Marks
1(a)	Any **two** from: • It has a **mechanical** structure/framework • It has **electrical** components // by example	2
1(b)	Any **two** from: e.g. • Employees don't need to lift heavy furniture • Employees can be protected from dangerous tasks • Employees can utilise their skills in other tasks • Employees don't need to perform repetitive/mundane tasks	2
1(c)	Any **one** from: e.g. • Expensive to **install/purchase/setup** • High ongoing costs/maintenance costs • May deskill the workforce • If they malfunction, production may stop	1

Question 2 · May/June 2023 Paper 12, original Q9 (a),(b) [5 marks]

Question	Answer	Marks
2(a)	**Three** from: • Rule base • Knowledge base • Interface	3
2(b)	Any **two** from: • It makes decisions • … by **applying** the rules/logic to the facts/knowledge … • … to provide a result/diagnosis	2

Total: 75 marks

Question 3 · Oct/Nov 2023 Paper 12, original Q10 (a),(b) [7 marks]

Question	Answer	Marks
3(a)	Any **one** from: – The ability to learn/adapt // machine learning abilities – The collection of data and the rules for using that data – The ability to reason // has problem solving abilities // makes predictions – Simulates intelligent/human behaviour – Analyses patterns	**1**
3(b)	Any **six** from: – It has an interface … – … used to input data/view output – It has a knowledge base – It has a rule base – It has an inference engine – Applies the rule base to/and the knowledge base to provide output/diagnosis/result/solution/decision – Decides what to ask next based on the data input	**6**

Total: 75 marks

Question 4 · Oct/Nov 2023 Paper 13, original Q10 (a),(b) [9 marks]

Question	Answer	Marks
4(a)	**One** mark for each correct term, in the correct order: – Knowledge base – Inference engine – Rule base // knowledge base – Knowledge base // rule base – Interface	**5**
4(b)	Any **four** from: e.g. – It is a form of artificial intelligence – Means it can adapt/change (its own processes) // It can edit its own algorithms – It can edit its own data – It can be trained – … this can be supervised/unsupervised – … meaning it can learn with/without human interaction – Analyses patterns and stores **successful**/**unsuccessful** results … – … to influence future decisions – (Supervised) means a user tells the system the input and output – (Unsupervised) means the system is given the input and needs to work out the output	**4**

Question 5 · May/June 2024 Paper 12, original Q10 (a),(b) [7 marks]

Question	Answer	Marks
5(a)	Any **two** from: • The **simulation** of intelligent behaviours by computers • A collection of data and the rules for using that data • Has the ability to reason • Has the ability to learn/adapt	2
5(b)	Any **five** from: • The user will enter data into the **interface** // The user will plug the car into an **interface** // The user is given questions using the **interface** • The **inference engine** will decide which questions to ask • … by using the previous answers given • The **inference engine** will decide on a diagnosis • … by **comparing the data/answers** entered to the **knowledge base** and **rule base** • … by calculating which option is most likely if there are multiple • The **interface** will output the **diagnosis/result** • The **explanation system** shows **how** the diagnosis was reached	5

Question 6 · Oct/Nov 2024 Paper 11, original Q9 (b) [1 marks]

Question	Answer	Marks
6(a)	Machine learning	1

Question 7 · Oct/Nov 2024 Paper 12, original Q8 (b),(c) [6 marks]

Question	Answer	Marks
7(a)	• Knowledge base • Rule base • Inference engine	3

7(b)	Any **three** from: For example: • It could **gather data** during **vacuuming** • … and adapt its **own processes** • … such as where obstacles are placed in the room • … such as where dirtier areas are • … such as a path through a room • … such as the shape of a room • … such as the most efficient route to vacuum a room • … so, it knows areas to avoid/concentrate on/use different cleaning tools	**3**

Question 8 · Feb/March 2025 Paper 12, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
8(a)	Any three from: • Collection of data • Rules for data • Ability to reason • Ability to adapt	**3**
8(b)	Any three from: Examples • **Changes** its data/rules • … according to feedback/results • If a result does not give a suitable response this is recorded • … next time the result is changed • … to provide more accurate/relevant results in future • Supervised/unsupervised learning	**3**

Total: 75 marks

Question 9 · May/June 2025 Paper 11, original Q8 (a),(b),(d),(e) [5 marks]

Question	Answer	Marks
9(a)	Any **one** from: • It has electrical components. // by example • It is programmable. • It can move.	1
9(b)	Any **one** from: • Permanent **instructions/software** that are programmed into/stored in the ROM. • **Instructions/software** that allow hardware to be controlled/managed. • **Instructions/software** that provides the operating system with a platform to run on.	1
9(c)	Any **two** from: Examples: • Workers don't need to lift **heavy** machinery • The workers are less likely to injure themselves • The workers don't need to be in any dangerous environments in the factory • The workers don't need to perform boring/repetitive tasks such as moving car parts • Workers can focus on having higher expertise skills • Create new jobs for workers to maintain the robots.	2
9(d)	Machine learning	1

Total: 75 marks

Question 10 · Oct/Nov 2025 Paper 13, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
10(a)	Any **three** from: • Simulation of intelligent behaviours by computers • Collection of data and rules for using that data • A computer that has the ability to learn • A computer that has the ability to adapt • A computer that has the ability to reason/make decisions	3
10(b)	Any **three** from For example: • It could use machine learning • It can collect data about the **players** actions • **Analyse/identify patterns** in the player's movements • … so it can learn how the player moves • … to help predict the next movement of the player • It can adapt its own movements to match the players // It can adapt its own processes • It can store successful/unsuccessful moves • It will learn the most efficient/optimal movements against a player	3

Total: 75 marks

Question 11 · May/June 2026 Paper 11, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks	Guidance
11(a)	Acoustic sensor	1	Mark the first answer given. Sound/noise sensor is TV. Microphone is NE.
11(b)	Any **two** from: • Proximity sensor • Infrared sensor • Light sensor • Magnetic field sensor	2	Mark the first answer given.
11(c)(i)	Any **one** from: • It has a mechanical structure/framework • It is programmable • It has the ability to move	1	Mark whole answer. MP1 – It is mechanical is TV MP2 – It can follow instructions is TV
11(c)(ii)	Any **three** from: • The microprocessor/it compares the received data/data sent to the stored value/preset value (of 10) • If the **data/value** received is **less than 10**, (the microprocessor) sends a **signal** to stop the robot • (The signal is sent) to an actuator that is used to stop the robot	3	MP2 – 'when an object/distance/robot is less than 10cm' is TV. MP3 – just reference to an actuator is NE here. Must be saying the actuator is what is used to trigger the action to stop the robot in some way.

11(d)	Any **four** from: For example: • It can use machine learning • … by adapting its own data/rules/processes • It can **create/makes** a (digital) map (of the room) // Records/collects data about the **shape/dimensions** of the room • … and **record/store** the **location** of objects • **GPS** may be used to record the location of objects • It can **update** its (digital) map • … to **record/store** when objects are **moved/removed/added** • It can store successful/unsuccessful routes/actions (that avoid objects)	**4**	Do not accept 'it can learn where objects are placed in the room'. This is given in the question.

Question 12 · May/June 2026 Paper 13, original Q6 (a),(b),(c),(d) [7 marks]

Question	Answer	Marks	Guidance
12	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct point
✗	Incorrect point
?	Unclear response
FT	Follow through
REP	Repetition
I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague

NAQ	Not answered question
Annotation	**Meaning**
λ	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

12(a)	Any **one** from: • It collects data • It applies rules to the data • It has the ability to reason	**1**	
12(b)	A program that has the ability to **automatically** adapt its **own** data and processes	**1**	
12(c)	**Two** from: For example: • It could **analyse** the **input** actions from the player • It could **analyse** past extra **characters** that performed well • Create/adapt the actions of an extra character based on past characters/gameplay	**2**	

12(d)	Any **three** from: • It takes/uses the data input by the user about the problem • … search for similar knowledge in the knowledge base • … applies it to the rule base (to find the solution) • … decide which question to ask next (if needed)	**3**	

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